NAUMAN DAWALATABAD

Research Scientist, Zoom Communications Inc., San Jose, CA, USA

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Research Scientist specializing in scalable Speech AI systems spanning multilingual ASR, conversational AI, speaker technologies, and efficient on-device deployment. Proven track record of translating research into production systems and open-source frameworks used across industry and academia.

RESEARCH INTERESTS

• Speech Recognition (ASR) • Speaker Diarization / Verification • Healthcare AI • Multimodal Audio+Text Learning

EXPERIENCE

Research Scientist

Zoom Communications, Inc.

May 2024 – Present | San Jose, CA, USA

- ***ASR for Voice Agents:*** Built production-grade S2S ASR systems for French and Hindi-English code-mixed conversational agents, leading end-to-end efforts from data acquisition/curation and augmentation to large-scale training, finetuning, and deployment, achieving >40% improvement over multilingual baselines and competing commercial systems.
- ***Word-Time Boundary Model:*** Built a lexicon-free, language-agnostic word boundary estimation model used in downstream applications like speaker diarization, speaker change detection, and end-of-unit detection for voice agents.
- ***Healthcare Clinical Notes:*** Built Zoom's first in-house healthcare ASR RNN-T system for clinicians. Adapted English ASR models using synthetic TTS data with LLM based personas, and contextual biasing. Improved performance >20%.
- ***Cross-functional collaborations:*** Led cross-functional collaboration across Data, legals, ASR, LLM, VAD, and eval teams.
- ***Open ASR Leaderboard:*** The data work included to our flagship EN model recently topped the Open ASR Leaderboard.

Postdoctoral Associate – MIT CSAIL

Massachusetts Institute of Technology

Dec 2021 – May 2024 | Cambridge, MA, USA

- **Supervisor:** Dr. James Glass
- ***Uncertainty-driven Pseudo-label Filtering and Model Calibration:*** Improved dropout-based uncertainty estimates for pseudo-label filtering. Demonstrated strong correlation between model calibration and the quality of pseudo-labels. [ICASSP'23]
- ***Speech+Text based Dementia Detection:*** Developed a trustworthy and explainable multimodal pipeline using speech and text signals for cognitive status estimation from long-form neuropsychological interviews. [EMNLP'22]
- ***Privacy-preserving Voice Conversion:*** Developed voice anonymization and conversion systems preserving speaker privacy while retaining linguistic content and prosodic characteristics.
- Selected as an IEEE ICASSP Rising Star in Signal Processing, 2023.

Lead Engineer – Bixby Speech & Audio Team

Samsung Research

Mar 2021 – Nov 2021 | Bangalore, India

- ***On-device ASR Model Compression:*** Developed progressive multi-stage knowledge distillation approaches for streaming two-pass RNN-T and Conformer-T ASR systems deployed on Samsung flagship and mid-range devices.
- Led key design ideas for Conformer-T model, resulting in more than 60% model size reduction and research publication.
- Received the *Samsung E-Spot Award 2021* for contributions to highly memory-efficient commercial ASR deployment.
- Successfully translated research prototypes into production systems for Samsung Galaxy devices. Work featured on the Samsung Research blog. This was used by large number of our Samsung phone customers.

Research Internship – SpeechBrain Team

MILA – Quebec AI Institute, University of Montreal

Feb 2020 – Feb 2021 | Montreal, Canada

- **Supervisors:** Prof. Yoshua Bengio and Dr. Mirco Ravanelli
- **Open-source:** One of the core developers of the SpeechBrain toolkit, ranking among the top contributors.
- Led ECAPA-based speaker diarization system achieving 30% rel. improvement over prior state-of-the-art baselines.
- Implemented and maintained (unit testing, integration testing, code reviews, demos, bug fixing, documentation, etc.) other core modules like spectral clustering, PLDA scoring, and x-vector speaker models. Released models on HF.

Mentorship, Leadership and Community

May 2017 – Present

- **Mentor/Leadership**: Mentored several interns and junior researchers in production-level development.
- Reviewed 60+ papers across major conferences and journals including ICASSP, Interspeech, ACL, EMNLP, IEEE TASLP, IEEE SPL, TPAMI, and Speech Communication.

TECHNICAL SKILLS

• PyTorch • TensorFlow • SpeechBrain • ESPNet • Kaldi • Python • C++

EDUCATION

Ph.D. and M.S. in Computer Science and Engineering

Indian Institute of Technology Madras

2014 – 2021 India

- **Supervisors:** Prof. C. Chandra Sekhar and Prof. Hema A. Murthy
- **Ph.D. Thesis:** "Speaker Diarization of Spontaneous Real-World Meeting Conversations."
  Research themes: (i) Test-time adaptation, (ii) Continual and incremental learning, (iii) Phoneme-aware segmentation strategies.
- **Other Projects:** Source separation, keyword spotting, factoid question answering, and CUDA FFT processing.
- Recipient of the *Institute Research Award 2021* for excellence in Ph.D. research.

B.E. in Information Technology

University of Mumbai

2008 – 2012 India

SELECTED PUBLICATIONS

- **N. Dawalatabad**, M. Ahangaran, C. Karjadi, L. Kourtis, P. Joung, V. Kolachalama, R. Au, J. Glass, "Differential Privacy Preserving Voice Conversion for Audio Health Data," *Alzheimer's & Dementia*, 2025. [link]
- **Nauman Dawalatabad**, Sameer Khurana, Antoine Laurent, James Glass, "On Unsupervised Uncertainty-Driven Speech Pseudo-Label Filtering and Model Calibration," *IEEE ICASSP*, 2023. [link]
- **Nauman Dawalatabad**, Yuan Gong, Sameer Khurana, Rhoda Au, James Glass, "Detecting Dementia from Long Neuropsychological Interviews," *Findings of ACL: EMNLP*, 2022. [link]
- **Nauman Dawalatabad**, Tushar Vatsal, Ashutosh Gupta, Sungsoo Kim, Shatrughan Singh, Dhananjaya Gowda, Chanwoo Kim, "Two-Pass End-to-End ASR Model Compression," *IEEE ASRU*, 2021. [link]
- **Nauman Dawalatabad**, Mirco Ravanelli, François Grondin, Jenthe Thienpondt, Brecht Desplanques, and Hwidong Na, "ECAPA-TDNN Embeddings for Speaker Diarization," *INTERSPEECH*, 2021. [paper]
- **Nauman Dawalatabad**, Srikanth Madikeri, C. Chandra Sekhar, Hema A. Murthy, "Novel Architectures for Unsupervised Information Bottleneck based Speaker Diarization of Meetings," *IEEE/ACM TASLP*, vol. 29, 2021. [paper]

INVITED TALKS AND RECOGNITION

- **Judge:** IEEE Signal Processing Cup Competition, IEEE ICASSP 2025.
- **Invited Research Talk:** "Speech Research in Healthcare at Zoom," AI Developers, Zoom Headquarters, San Jose, 2024.
- **Alumni Guest Invited Talk:** "Robust Speech Recognition," IIT Madras, 2023.
- **Guest Lecturer:** "Speaker Recognition and Diarization," MIT 6.345, MIT CSAIL, 2022.
- **IEEE ICASSP Rising Star in Signal Processing,** 2023.

IN NEWS ARTICLES

- **CNBC News**: Voice AI Cloning. [Interview Link]
- **Boston Globe**: Speech voice converstion [Interview Link]
- **MIT News**: Speech Anonymization [Interview Link] [Video]
- **IEEE Spectrum AI News**: Detect Deep Audio fakes. [Interview Link]